

Syllabus

Spring 2026.

Yurii Shylnov.

Quantum Theory II. Phys-510, section 01.

- Sessions:** Monday/Wednesday 1:50 pm-3:05 pm. Classroom: RE-103.
- Instructor:** Prof. Yurii Shylnov.
- Office:** PS-396.
- Phone/ E-mail:** Phone: (312) 567-3443. E-mail: shylnov@illinoistech.edu
- Office hours:** MW 12:50-1:50pm or by appointment.
Although sometimes I might have a meeting during a lunch-time break.
- Textbooks:**
1. **Bipin Desai. Quantum Mechanics with Basic Field Theory. Cambridge University Press, 2009. 838 p. ISBN: 978-0521877602.**
 2. **J. J. Sakurai, Jim Napolitano. Modern Quantum Mechanics. Third Edition. Cambridge University Press, 2020. 566 p. ISBN: 9781108473224.**
- Objectives:**
1. Continue to learn the second quantization formalism and its applications to various physical systems.
 2. Develop the quantum theory of scattering from the first principles and solve problems using this theory.
 3. Explore the relativistic quantum mechanics, its new concepts and fundamental results.
 4. Grasp key concepts of the quantum field theory, e.g., gauge invariance, spontaneous symmetry breaking mechanism etc.
- Homework:** You will mostly receive your homework every other week. All homework problems are to be done systematically and legibly. I am going to collect it and grade one problem selected randomly.
- Attendance:** Attendance will be taken into account in borderline cases.
- Project:** I think about a quantum methods problem of a good level but still doable. The key component is that you should be interested in exploring the problem. This is your second semester, and I expect your projects to be deeper, better organized and structured.

The title and abstract should be submitted by March 11th. The first draft of the paper should be submitted by April 15th. A final version should be submitted and discussed by April 27th. You are going to make presentation of your project in the class during the last week of the semester. Probably, we will add one more session to listen to every student.

Tests:	There will be three tests on February 23 rd , March 30 th , and April 20 th . No final exam.
Grading:	Homework -20%, Tests - 60%, Project-20%.
Course Grade Determination:	A 90-100%, B 80-89.9%, C 70-79.9%, D 60-69.9%, E below 60%.
Course Description:	1. Second quantization. 2. Identical particles. Multiparticle systems. 3. Bogolyubov transformations. Quantum chains. SYK model. 4. Rotations and angular momentum. Angular momenta algebra. 5. Scattering theory. Effective cross-section, partial phases, Born approximation. 6. O(3) group applications in quantum mechanics. Spin and spinors. Pauli equation. 7. Relativistic quantum mechanics. Klein-Gordon equation. Multiparticle interpretation. 8. Dirac equation. γ-matrices and their algebra. 9. Lorentz group, its representations, Lorentz spinors. Bispinors. 10. Dirac equation solution for a free particle. 11. Gauge invariance in QM. Spinor and scalar electrodynamics. 12. Spontaneous symmetry breaking. Higgs mechanism.
Academic honesty:	It is presumed that you will do your own work on the homework and tests. Discussing homework problems with others is encouraged but submitting work as your own, which is copied or paraphrased from someone else is not permitted. Cheating includes, but not limited to, illegal collaboration, copying, using materials not permitted on tests, and aiding others on tests. Anyone found cheating will not be permitted to withdraw and will receive a grade of E for the course. Your academic dean will be informed and a statement will be placed in your permanent file. For more details, see the student handbook: https://www.iit.edu/student-affairs/student-handbook/fine-print/code-academic-honesty
Classroom environment:	Please, no cellular phones, iPods, iPads etc. in the class. I will be very much indebted to you if you do not eat, talk, text and so on in the classroom. It distracts everybody. You are allowed to use your notebooks, tablets etc. to take notes but nothing else. The class format is going to be quite traditional. Lectures and discussions will take the most time.
Americans with Disabilities Act Policy Statement:	Reasonable accommodation will be provided for students with documented disabilities. Contact the Center for Disability Resources to receive necessary support and help. The Center for Disability Resources (CDR) is located in 10 W. 35th St., Suite 3F3-1, phone: 312.567.5744 , e-mail: disabilities@illinoistech.edu
Title IX Compliance:	Title IX and our school policy prohibits discrimination on the basis of sex as well as all forms of sexual misconduct or harassment. I am always concerned about each and every student's well-being and ready to help. However, I am also required by the law to report any such incident to an Illinois Tech's Title IX coordinator. If you have experienced such an incident, please consult the following link for a list of resources available to you: https://web.iit.edu/hea-compliance/sexual-harassment-and-misconduct-prevention/title-ix-illinois-tech/resources.

Some Important Dates

January 12	Classes Begin.
January 19	MLK Day. No Classes.
January 23	Make up for Jan 19 Classes
February 23	Test # 1.
March 13	Midterm Grades Due.
March 15-23	Spring Break. No Classes.
March 30	Test # 2.
April 20	Test # 3.
April 27	Project Due
April 29	Last Day of Classes.
May 13	Final Grades Due.